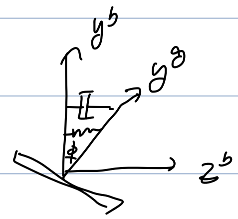
mech gyro

$$\begin{cases} \underline{h}^i_{ig} = I_g \underline{\omega}^i_{ig} \\ \dfrac{d}{dt}\left(\underline{h}^b_{ig}\right) = \sum \underline{T}^b_{ext} \end{cases}$$

$$\dot{h}^b_{ig} = \dot{C}^b_i \underline{h}^i_{ig} + C^b_i \dot{\underline{h}}^i_{ig}$$

$$= -\Omega^b_{ib} C^b_i \underline{h}^i_{ig} + C^b_i \dot{\underline{h}}^i_{ig}$$



$$\Rightarrow C^b_i \dot{\underline{h}}^i_{ig} = \dot{\underline{h}}^b_{ig} + \Omega^b_{ib} \underline{h}^b_{ig} = \sum \underline{T}^b_{ext}$$

$$T^b_x = -c\dot{\phi} - k\phi$$

$$\underline{\omega}^b_{ib} = [p, q, r]^T, \quad \underline{\omega}^b_{bg} = [\dot{\phi}, \omega_s, 0]^T$$

$$\underline{\omega}^b_{ig} = \underline{\omega}^b_{ib} + \underline{\omega}^b_{bg} = \begin{bmatrix} p + \dot{\phi} \\ q + \omega_s \\ r \end{bmatrix}$$

$$\underline{h}^b_{ig} = I_g \underline{\omega}^b_{ig} = \begin{bmatrix} I_{xx}(p + \dot{\phi}) \\ I_{yy}(q + \omega_s) \\ I_{zz} r \end{bmatrix}$$

$$\Omega^b_{ib} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}$$

$$\Omega^b_{ib} \underline{h}^b_{ig} = \begin{bmatrix} -r I_{yy}(q + \omega_s) + q I_{zz} r \\ r I_{xx}(p + \dot{\phi}) - p I_{zz} r \\ -q I_{xx}(p + \dot{\phi}) + p I_{yy}(q + \omega_s) \end{bmatrix}$$

$$\dot{h}^b_{ig} = \begin{bmatrix} I_{xx}(\dot{p} + \ddot{\phi}) \\ I_{yy} \dot{q} \\ I_{zz} \dot{r} \end{bmatrix}$$

$$x: \quad I_{xx}(\dot{p} + \ddot{\phi}) - r I_{yy}(q + \omega_s) + q I_{zz} r = -c\dot{\phi} - k\phi \quad (\omega_s \gg q)$$

$$\Rightarrow \ddot{\phi} + \frac{c}{I_{xx}} \dot{\phi} + \frac{k}{I_{xx}} \phi = \frac{(I_{yy}\omega_s - I_{zz} q) r}{I_{xx}} - \dot{p}$$

$$\approx \frac{I_{yy}\omega_s}{I_{xx}} r$$